

Gandharva Appagere

EXPERIMENTAL PHYSICIST • DETECTORS, DATA & AI

PhD candidate in experimental nuclear & hadron physics, Stockholm University

Stockholm

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PROFILE

Experimental physicist (PhD candidate, Stockholm University) with five years of hands-on work on a large accelerator experiment at GSI/FAIR in Germany. Main analysis responsibility for a rare-process measurement: detector operation, calibration-aware particle identification, Monte Carlo simulation, statistical inference with full uncertainty treatment, and machine-learning classification. Applies modern AI tools — ML classifiers and AI coding assistants — on top of deep instrumental expertise to build reproducible, automated analysis pipelines on HPC clusters. Earlier industry experience in automotive radar algorithms (Robert Bosch) and electromagnetic engineering. Experienced leader in university governance.

Completing my PhD in 2027 and open to industry R&D and data roles. Sectors of interest: Sensing & defence systems • Measurement technology & metrology • Medical technology & radiation • Autonomous systems • Applied R&D • Quantitative analysis

FROM THE LAB TO INDUSTRY

Detectors & sensor systems

Specify, integrate and debug multi-sensor systems; reason from raw sensor behaviour to system performance.

Signal from noise: statistics & uncertainty

Robust inference, detection and estimation with honest uncertainty budgets for decisions that depend on noisy data.

AI/ML & AI-assisted workflows

Apply ML to physical and sensor data with domain understanding, and use modern AI tools to ship analysis software faster.

Measurement, calibration & instrumentation

Design calibration strategies, characterise instruments and keep measurements traceable and reproducible.

Simulation & Monte Carlo

Build digital twins and synthetic data, run sensitivity and feasibility studies before hardware exists.

Large-scale data & HPC

Engineer data pipelines that scale from a notebook to a compute cluster, with reproducible, reviewable code.

EXPERIENCE

Doctoral researcher (PhD candidate)

Stockholm University • HADES & CBM collaborations, GSI/FAIR

Aug 2021 - present

Stockholm, Sweden & Darmstadt, Germany

Experimental hadron physics on the HADES spectrometer at GSI and simulation studies for the CBM experiment at FAIR.

- Main analysis responsibility in the HADES collaboration for the $\Lambda\Lambda$ channel in proton-proton collisions: event selection, reconstruction, simulation, signal extraction and systematic uncertainties.
- Took detector shifts during the five-week February 2022 proton beam time and a 2025 pion-beam run; worked with drift-chamber online quality assurance.
- Developed a relative time-of-flight particle-identification method that compensates start-detector timing problems, used with kinematic fitting and 2D template likelihood fits.
- Trained multiclass machine-learning classifiers (ROOT TMVA) to separate $\Lambda\Lambda$ signal from background.
- Built a staged (S0–S3) analysis pipeline with automated systematic variations on the GSI batch farm (Slurm), developed with AI coding assistants (Cursor, LLMs).
- Wrote an event generator and detector-response folding for CBM feasibility studies of $\Lambda\Lambda$ cusps; contributed to the QCD at FAIR white paper.
- Hosted and minuted the weekly meetings of the HADES Λ -Reconstruction Task Force and Kinematic-Fitting working group (2022–23).

C++ • ROOT • Python • TMVA • Slurm • PLUTO • GEANT • KinFit • Git

Electric Propulsion Engineer

KTH Hyperloop team

Sep 2018 - 2019

Stockholm, Sweden

Linear induction motor for the SpaceX Hyperloop Pod Competition 2019.

- Designed a double-sided linear induction motor for the pod's propulsion; electromagnetic finite-element simulations in COMSOL and ANSYS Maxwell, with MATLAB.

COMSOL • ANSYS Maxwell • MATLAB

Associate Software Engineer

Robert Bosch Engineering and Business Solutions (RBEI)

Aug 2017 - Jul 2018

Bengaluru, India

- Worked on signal-processing algorithms for automotive radar sensors used in driver-assistance systems, in an industrial software-engineering environment.

EARLIER PROJECTS & INTERNSHIPS

Intern, Network Engineering Facility, AT&T, Bengaluru. VPN order-management processes in a network-engineering organisation.

Jan-Jul 2017

Hybrid levitation for Hyperloop — engineering project, BMS College of Engineering, Bengaluru. Design of air-bearing levitation combined with linear-induction-motor propulsion for a Hyperloop concept. Aug 2016 – Jun 2017

Intern, Quality Assurance division, ISRO — Laboratory for Electro-Optics Systems (LEOS), Bengaluru. Circuit analysis of star, Earth and Sun sensors for the ASTROSAT and Chandrayaan-2 missions. Jul 2015

EDUCATION

PhD, Experimental Nuclear & Hadron Physics • *Stockholm University* 2021–2027 (expected)

Licentiate of Philosophy (Fil.lic.), Physics • *Stockholm University* 2021–2024

Thesis: “ Λ - Λ hyperons: the road to estimate the production cross-section in proton-proton collisions”

MSc, Electromagnetics, Fusion and Space Engineering • *KTH Royal Institute of Technology* 2018–2020

BE, Electronics and Communication Engineering • *BMS College of Engineering* 2013–2017

CORE SKILLS

Data analysis, ML & AI Statistical inference (maximum-likelihood & template fits, least squares, sideband methods) • Uncertainty quantification (systematic variations, correlated averages) • Kinematic (constrained) fitting (vertex and multi-constraint fits) • Machine-learning classification (multiclass classifiers in ROOT TMVA) • AI-assisted development (Cursor and LLM coding assistants for building and automating pipelines)

Programming & software Python (NumPy, SciPy, pandas, matplotlib, uproot, iminuit, Jupyter) • C++ & ROOT (analysis frameworks, PyROOT, RooFit) • Git / GitLab

Simulation & modelling Monte Carlo event generation (PLUTO, UrQMD, own generator with final-state interactions) • Detector simulation (GEANT3-based HGeant, CbmRoot) • Finite-element electromagnetics (COMSOL, ANSYS Maxwell)

Detectors & instrumentation Tracking & timing detectors (drift chambers, straw tubes, RPC, LGAD start detector, time of flight) • Gamma spectroscopy (HPGe detectors, scintillators, neutron activation) • Beam-time operations & online data-quality monitoring • Radiation safety & radioactive-source handling • Automotive radar signal processing

Computing & HPC HPC batch computing (Slurm on the GSI Virgo cluster) • Containers & software distribution (Singularity/Apptainer, CVMFS) • Large datasets (~155 TB raw data, Lustre, ROOT trees, HDF5)

SELECTED PUBLICATIONS & TALKS

G. Appagere (for the HADES and CBM Collaborations), “Investigating Double-Strange Systems at FAIR: $\Lambda\Lambda$ interactions and neutron star cores”, *JPS Conference Proceedings — Proceedings of the 15th International Conference on Hypernuclear and Strange Particle Physics (HYP2025)*, Tokyo. **IN PRESS** 2026

J. G. Messchendorp, F. Nerling, ..., **G. Appagere**, et al., “**Hadron Physics Opportunities at FAIR**”, *arXiv:2512.15986 [hep-ex]*. **PREPRINT** 2025

9 conference talks, including NSTAR 2026 (Seville), 15th Nordic Meeting on Nuclear Physics (Visby, 2026), DPG Spring Meeting 2026 (Erlangen) and HYP2025 (Tokyo); plus 20 presentations at internal HADES and CBM collaboration meetings. 2022–2026

LEADERSHIP & RESPONSIBILITY

Chair (2025–2026), **Vice-Chair** (2023–2025), Fysikum PhD Council, Stockholm University. 2023–2026

Vice-Chair, Central Doctoral Student Council (CDR), Stockholm University. 2024–2025

Vice-Chair, Natural Sciences Faculty Doctoral Student Council (NFR), Stockholm University. 2024

Doctoral-student representative, Fysikum Department Board and committees (Research Education, Technical Division, Computer, Work Environment). 2023–2026

Lab responsible, Nuclear physics student laboratory, Fysikum. Responsible for HPGe detectors, scintillators and the radioactive-source inventory; lab rules and safety, a source booking and record system, and procurement of a proportional counter. 2021–2025

University teaching, Stockholm University. Laboratory supervision, exercise classes and grading in 6 physics courses; co-supervision of a Bachelor’s thesis in Medical Radiation Physics. 2021–2025

AWARDS & LANGUAGES

Awards SAE E-Baja 2017 — 3rd place in India, design review • IEEE Darrel Chong Student Activity Award (2016) • SAE Aero Design West 2015 — 13th worldwide (team captain)

Grants 4 grants and scholarships for research travel and experiments (Stockholm University, 2022–2026)

Languages English: Fluent — professional working proficiency • Swedish: Basic (A2, studying)